

several development, certification and coexistence tests for the same.”

A vehicle radar test system is a pre-configured collection of National Instruments (NI) products and intellectual properties specifically designed for testing radar technology in 76GHz - 81GHz range, from validation through high-volume production.

It covers individual radar sensors all the way to integrated ADASes.

One key advantage of a vehicle radar test system is its ability to provide dynamic obstacle generation and comprehensive RF characteristic measurements on top of standard simulators that only work for functional behavioural tests. NI provides solutions to test ADAS, V2X, infotainment, powertrain, and hardware and software as a fully integrated system.

Testing wireless devices

Broadcast technologies keep car occupants connected and entertained. Today, the vehicle incorporates digital radio and video broadcasting. Madhukar Tripathi, head - marcom and optical business, Anritsu India Pvt Ltd, says, “Anritsu automotive test solutions validate the operation of connected car communication systems. These allow tests at all stages

of product development, from R&D to production, to provide dedicated solutions and universal wireless testers. This ensures that the products comply with latest industry specifications.

“Anritsu also provides test solutions for electromagnetic interference (EMI), wireless connectivity in vehicle networks, safety and driver aids, and intelligent transport systems.”

Rajesh Gupta, general manager, TUV Rheinland India Pvt Ltd, says, “Our test facilities support complete electromagnetic compatibility (EMC) and wireless testing, according to Automatic Identification System (AIS) requirements, and European and other international standards. These also take care of original equipment manufacturer (OEM)-specific requirements, such as immunity testing up to 200V/m using strip line test method.

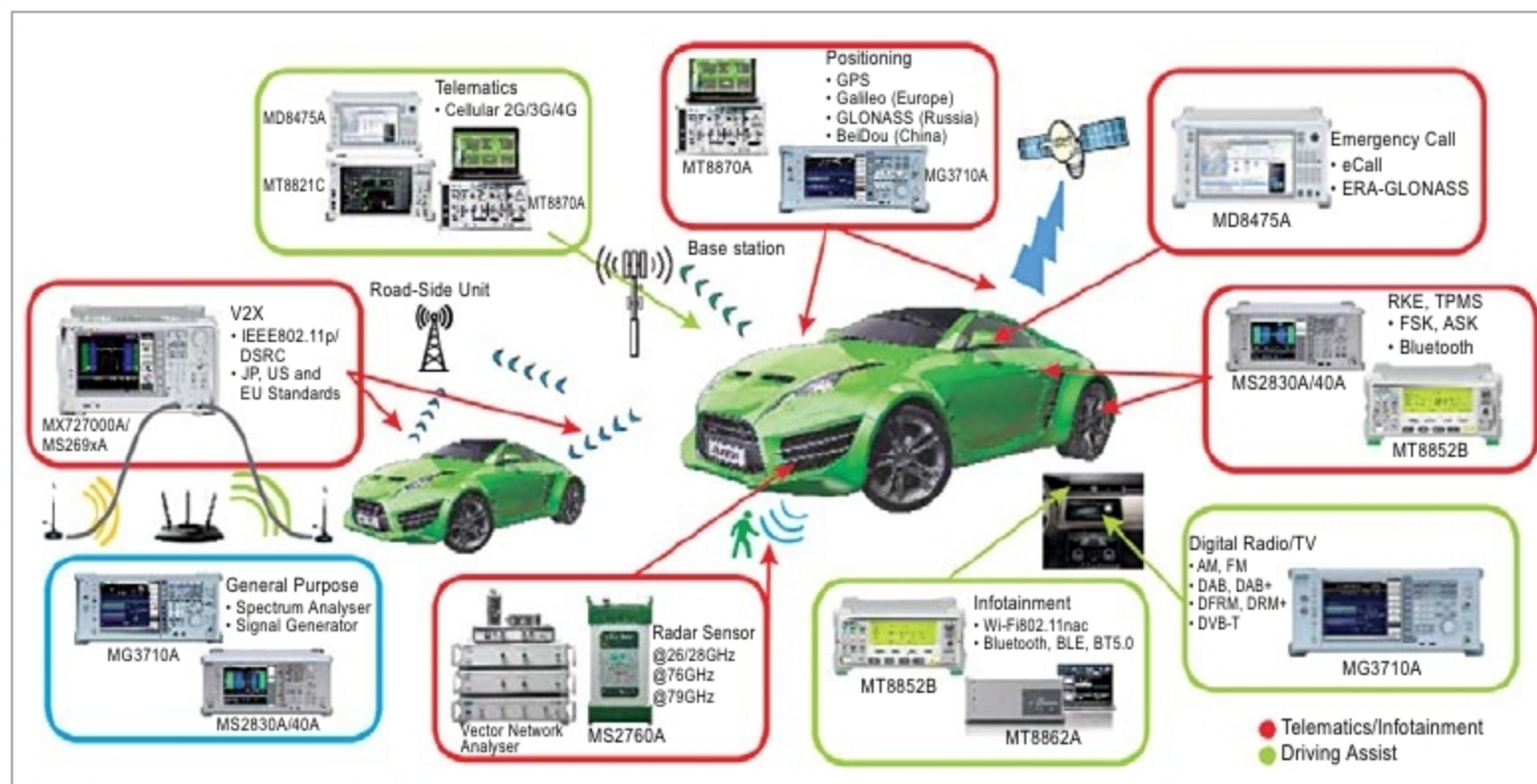


Fig. 2: High-performance automotive test solutions (Credit: Anritsu)

“Our expansion plan includes whole vehicle EMC testing with dynamo, complete OEM-specific requirements including electrical, reliability, wireless antenna performance and cellular technology-related testing services.”

Testing environmental conditions

Velocity of air used during painting is extremely critical and should be maintained at a speed of 1m/s to 1.5m/s. CO₂ levels in the paint shop must be measured to ensure healthy working conditions. Air coming from the compressor house to the paint shop must be clean, dry and moisture-less. Dew that may seep in the air must be monitored before use. Else, it can directly hamper paint quality.

As per international studies, maximum 20 per cent air leakage is permitted, but instruments can help reduce these losses. Reducing leakages helps in peak load management of air.

Before applying paint on the vehicle body, it must be kept in a separate chamber, where it is completely dried. This is known as the baking process. Efficiency of the heating dryer must be measured periodically.

Kalidas Bhangare, managing director, Testo India Pvt Ltd, says, “Testo 440 measures multiple param-

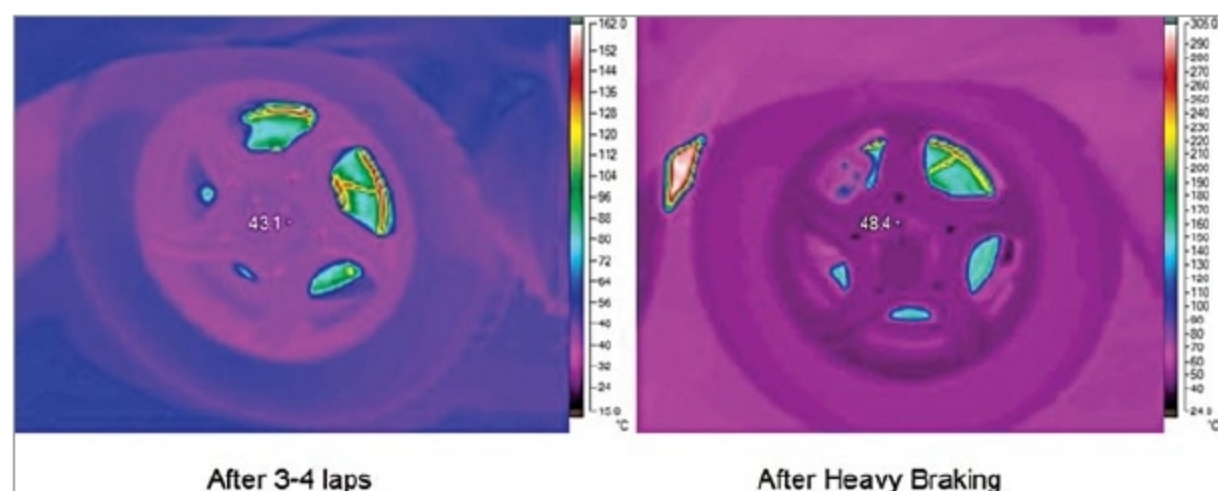


Fig. 3: Thermal imaging of electronic components (Credit: Testo India Pvt Ltd)